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ELECTRIC STORAGE DEVICE, AND VEHICLE

Abstract

An electric storage device includes: a first case that houses a first cell group; and a second case that houses a second cell group. Each of the first cell group and the second cell group includes a plurality of electric storage cells that is connected in a first direction. The first case includes an opening portion at one end in a second direction orthogonal to the first direction. The first case and the second case are disposed such that the second case closes the opening portion of the first case.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application claims priority to Japanese Patent Application No. 2024-025487 filed on Feb. 22, 2024, incorporated herein by reference in its entirety.
BACKGROUND
1. Technical Field
[0002] The present disclosure relates to an electric storage device that includes a plurality of electric storage cells, and vehicle.
2. Description of Related Art
[0003] Japanese Unexamined Patent Application Publication (Translation of PCT application) No. 2023-502457 (JP 2023-502457 A) discloses a battery (electric storage device) having a rectangular parallelepiped shape in which a length L is 400 mm to 2500 mm and the ratio (L/H) of the length L to a width H is 4 to 21.
SUMMARY
[0004] In the electric storage device described in JP 2023-502457 A, a plurality of electrode sets (electric storage cells) that is connected in series and is arrayed on a line is disposed in a case (housing). Such an electric storage device includes an aggregation (cell group) of a plurality of electric storage cells that is electrically connected.
[0005] In the production of the above electric storage device, the cell group needs to be put in the case. For example, it is possible that the cell group is put in the case from an opening portion provided on the case, the opening portion of the case is subsequently closed by a lid member, and the case body and the lid member are joined. However, when the opening portion of the case is small, it is difficult to put the cell group in the case. Conversely, when the opening portion of the case is large, a large lid member (or many lid members) is necessary for closing the opening portion. The increase in the dimensions or number of components sometimes makes it difficult to produce the electric storage device, or causes the rise in production cost.
[0006] The present disclosure has been made for solving the above problem, and has an object to provide an electric storage device that makes it possible to restrain the increase in the dimensions or number of components and that is easily produced, and a vehicle.
[0007] As a mode of the present disclosure, the following electric storage device is provided.
[0008] (First Item) The electric storage device includes: a first case that houses a first cell group; and a second case that houses a second cell group. Each of the first cell group and the second cell group includes a plurality of electric storage cells that is connected in a first direction. The first case includes an opening portion at one end in a second direction orthogonal to the first direction. The first case and the second case are disposed such that the second case closes the opening portion of the first case.
[0009] In the above configuration, the opening portion of the first case that houses the first cell group is closed by the second case that houses another cell group (second cell group), and therefore, it is not necessary to separately provide a lid member for closing the opening portion of the first case. Thereby, the increase in the number of components is restrained. At least one face of the second case is formed so as to have such a size that the second cell group can be housed. Therefore, even when an opening portion having a size that allows the first cell group to easily pass is formed on the first case, the opening portion of the first case can be closed by the second case. In the above configuration, it is possible to provide an electric storage device that makes it easy to put the cell group in the case, that is, an electric storage device that is easily produced.
[0010] (Second Item) In the electric storage device according to the first item, each of the first case and the second case has a rectangular parallelepiped shape in which an opening portion is provided

at one end in the second direction. Each of the first case and the second case includes a first face and a second face that are positioned at both ends in the first direction, and a third face, a fourth face, and a fifth face that extend in the first direction. The third face is positioned on the opposite side of the opening portion in the second direction. The opening portion of the first case has a larger area than each of the first face and the second face of the first case. The third face of the second case closes the opening portion of the first case.
[0011] In the above configuration, the opening portion having a larger area (opening area) than each of the two end faces (the first face and the second face) in the first direction is formed on the first case. A larger opening portion is formed on the first case, compared to a configuration in which an opening portion is formed on one of the end faces of the first case, and therefore, the first cell group is easily put in the first case through the opening portion. Further, the case has a rectangular parallelepiped shape, which is a simple shape, and therefore, is easily produced. The first case and the second case have an identical shape or similar shapes as described above, and this also contributes to the reduction in production cost. In the above configuration, the electric storage device is easily produced, and the production cost is easily reduced.
[0012] (Third Item) In the electric storage device according to the second item, the first case and the second case are joined such that the first case is sealed in a state where the first case houses the first cell group. The electric storage device further includes a lid member that closes the opening portion of the second case such that the second case is sealed in a state where the second case houses the second cell group.
[0013] In the above configuration, each case can be sealed in the state where the case houses the cell group.
[0014] (Fourth Item) The electric storage device according to any one of the first to third items, further includes a mechanism that generates force by which the second case is pressed against the first case.
[0015] In the above configuration, it is possible to lessen (or eliminate) the gap between the opening portion of the first case that houses the first cell group and the second case.
[0016] As a mode according to a second aspect of the present disclosure, the following vehicle is provided.
[0017] (Fifth Item) The vehicle includes the electric storage device according to any one of the first to fourth items.
[0018] The above vehicle includes an electric storage device having a configuration that makes it possible to restrain the increase in the dimensions or number of components and that is easily produced.
[0019] With the present disclosure, it is possible to provide the electric storage device that makes it possible to restrain the increase in the dimensions or number of components and that is more easily produced, and the vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:
[0021] FIG. **1** is a diagram for describing the configuration of an electric storage device according to an embodiment of the present disclosure;
[0022] FIG. **2** is a diagram for describing the configuration of each cell coupling body shown in FIG. **1**;
[0023] FIG. **3** is a sectional view taken along line III-III in FIG. **1**;

[0024] FIG. **4** is a diagram for describing operations and effects of the electric storage device according to the embodiment of the present disclosure;
[0025] FIG. **5** is a diagram showing a first modification of the electric storage device shown in FIG. **1**; and
[0026] FIG. **6** is a diagram showing a second modification of the electric storage device shown in FIG. **1**.

DETAILED DESCRIPTION OF EMBODIMENTS

[0027] An embodiment of the present disclosure will be described in detail with reference to the drawings. In the figures, identical or corresponding portions are denoted by identical reference characters, and descriptions thereof are not repeated. In each figure that is used below, an X-axis, a Y-axis, and a Z-axis are orthogonal to each other, the X-axis indicates a first in-plane direction (for example, a length direction) of the electric storage device, the Y-axis indicates a second in-plane direction (for example, a width direction) of the electric storage device, and the Z-axis indicates a height direction of the electric storage device. Hereinafter, the directions of arrows of the X-axis, the Y-axis, and the Z-axis are expressed with "+", and the opposite directions are expressed with "−".
[0028] FIG. **1** is a diagram for describing the configuration of an electric storage device according to the embodiment. The electric storage device **100** according to the embodiment is configured to be capable of storing electricity. The electric storage device **100** includes a plurality of electric storage cells each of which functions as a secondary battery. Details will be described later. Examples of the secondary battery include a lithium-ion battery, a nickel-hydrogen battery, and a sodium-ion battery. Examples of the lithium-ion battery include an LFP battery in which lithium ferrous phosphate is employed as a positive electrode active material, and a ternary battery in which nickel-manganese-cobalt (NMC) is employed as a positive electrode active material. The type of the secondary battery may be a liquid secondary battery or an all-solid-state secondary battery. The electric storage device **100** may include only the same kind of electric storage cells (for example, only LFP batteries), or may include different kinds of electric storage cells (for example, LFP batteries and ternary batteries).
[0029] The interiors of cases as viewed from the +Z-side are shown on the left side of FIG. **1**. The electric storage device **100** includes a case **301** (first case) that houses a cell group G.sub.A (first cell group), a case **302** (second case) that houses a cell group G.sub.B (second cell group), and a case **303** that houses a biasing device **50**. Each of the cases **301** to **303** has a rectangular parallelepiped shape in which an opening portion is formed on the +Y-side. Each corner of the cases **301** to **303** may be beveled (for example, by rounding process). The X-direction corresponds to the longitudinal direction of each of the cases **301** to **303**. In the embodiment, the X-direction and the Y-direction correspond to examples of the "first direction" and the "second direction" according to the present disclosure, respectively.
[0030] For example, metal can be employed as the material that composes each of the cases **301** to **303**. For example, each of the cases **301** to **303** is a case made of aluminum. However, without being limited to this, each of the cases **301** to **303** may be a case made of resin. Further, a different material may be employed for each of sites (for example, later-described faces F**1** to F**5**) of the cases. In the embodiment, the case **301**, the case **302**, and the case **303** have the same dimensions and the same shape. Therefore, as a representative, only the dimensions and shape of the case **301** will be described below in detail.
[0031] The case **301** includes faces F**1**, F**2** that are positioned at both ends in the X-direction, and faces F**3** to F**5** that extend in the X-direction. The face F**1** (first face) covers a +X-side end portion of the cell group G.sub.A that is housed in the case **301**. The face F**2** (second face) covers a −X-side end portion of the cell group G.sub.A that is housed in the case **301**. The case **301** includes the face F**3** (third face) at one end (−Y-side) in the Y-direction, and includes an opening portion at the other end (+Y-side) in the Y-direction. The face F**3** is positioned on the opposite side of the opening

portion in the Y-direction. In the embodiment, the opening portion is wholly formed on the face on the opposite side of the face F**3**, but the opening portion may be partially formed. The face F**4** (fourth face) and the face F**5** (fifth face) correspond to a pair of opposed faces that are opposed to each other in the Z-direction. Each of the faces F**1** to F**5** correspond to a plate-shaped portion that constitute the case **301**. However, the shape of each of the cases **301** to **303** is not limited to the above shape.

[0032] Each of the opening portion and the face F**3** on the opposite side of the opening portion has a larger area than each of the faces F**1**, F**2**. The length (the dimension in the X-direction) of the case **301** is longer than the width (the dimension in the Y-direction) of the case **301**. The length of the case **301** may be 250 mm or more and 5000 mm or less, and for example, is about 1000 mm. The width of the case **301** may be 10 mm or more and 1250 mm or less, and for example, is about 50 mm. The proportion of the length of the case **301** to the width of the case **301** may be 4 or more and 25 or less. The height (the dimension in the Z-direction) of the case **301** may be 10 mm or more and 1250 mm or less, and for example, is about 100 mm. The dimensions of each of the cases **301** to **303** are not limited to the above dimensions.

[0033] In the electric storage device **100**, as shown in FIG. **1**, the face F**3** of the case **302** closes the opening portion of the case **301**. The case **301** and the case **302** are joined such that the case **301** is sealed in a state where the case **301** houses the cell group G.sub.A. Further, the face F**3** of the case **303** closes the opening portion of the case **302**. The case **302** and the case **303** are joined such that the case **302** is sealed in a state where the case **302** houses the cell group G.sub.B. The case **303** functions as a lid member that closes the opening portion of the case **302**.

[0034] The biasing device **50** housed in the case **303** includes a mechanism that generates the force (the load to the −Y-side) by which the face F**3** of the case **303** is pressed against the case **302**. By the transmission of the force (load), the face F**3** of the case **302** is further pressed against the case **301**. The biasing device **50** may generate biasing force, and may give the force to the case **302** using the biasing force. The biasing device **50** may include a mechanism that mechanically generates the biasing force (for example, a mechanism that uses a spring). Further, the biasing device **50** may include a mechanism that generates the biasing force by electric power (for example, a mechanism that uses an electric actuator). Since the face F**3** (plate-shaped portion) of the case **303** is pressed against the case **302**, it is possible to lessen (or eliminate) the gap between the opening portion of the case **302** and the face F**3** of the case **303**. Further, since the face F**3** (plate-shaped portion) of the case **302** is pressed against the case **301**, it is possible to lessen (or eliminate) the gap between the opening portion of the case **301** and the face F**3** of the case **302**. Since the opening portion of each of the cases **301**, **302** is closed by another case (case **302** or **303**) in this way, the airtightness of the cases **301**, **302** is enhanced. In order that the face pressure due to the above force (load) is uniformly given to the cases **301**, **302**, an elastic body and/or an adhesive agent may be provided between the case **301** and the case **302** and between the case **302** and the case **303**.

[0035] In the electric storage device **100**, the cases **301** to **303** (three housings each of which includes the opening portion on the +Y-side) are stacked in the Y-direction. Thereby, the volumetric efficiency of the electric storage device **100** is enhanced. Further, the opening portion of the case **303** that is at the outermost position on the +Y-side among the cases **301** to **303** is not closed and is opened. Thereby, the heat dissipation of a housing portion (including the biasing device **50**) of the case **303** is enhanced. However, without being limited to this structure, the opening portion of the case **303** may be closed by an unillustrated lid member.

[0036] On the face F**1** of each of the cases **301**, **302**, components for managing the housed cell group are provided. The face F**1** of the case **301** and the face F**1** of the case **302** have the same structure, and therefore, as a representative, an enlarged view of the face F**1** of the case **301** is shown on the right side in FIG. **1**. A sealing hole **310**, an external terminal **320**, and a connector **330** are provided on the face F**1** of each of the cases **301**, **302**. The sealing hole **310** may be a pressure regulation hole that regulates the pressure in the case. For example, the sealing hole **310** has a seal structure that is constituted by a metal cap (case outer side) and a seal member (case inner side). By this seal structure, the airtightness of the case is secured, and when the pressure in the case exceeds a predetermined level, gas is emitted to the exterior of the case through the sealing hole **310**. The external terminal **320** includes an electrode tab **321** that is joined to a first connection terminal of the cell group (for example, by laser welding), and an electrode tab **322** that is joined to a second connection terminal of the cell group (for example, by laser welding). For example, at the external terminal **320** provided on the face F**1** of the case **301**, the electrode tab **321** is joined to a connection terminal T**1** of the cell group G.sub.A, and the electrode tab **322** is joined to a connection terminal T**2** of the cell group G.sub.A. Further, at the external terminal **320** provided on the face F**1** of the case **302**, the electrode tab **321** is joined to a connection terminal T**3** of the cell group G.sub.B, and the electrode tab **322** is joined to a connection terminal T**4** of the cell group G.sub.B. Each of the electrode tabs **321**, **322** may have an insulating seal structure with ceramic, around an electrode, for example. In the embodiment, the electrode tabs **321**, **322** function as a negative electrode tab and a positive electrode tab, respectively. However, without being limited to this, polarities may be reversed such that the electrode tab **322** is the negative electrode tab and the electrode tab **321** is the positive electrode tab. For example, the connector **330** includes an output terminal that outputs a detection signal indicating the state (for example, the temperature of each electric storage cell) in the case that is detected by one or more sensors in the case, to the exterior of the case, and an input terminal that inputs a control signal from the exterior of the case to one or more devices in the case. For example, a temperature sensor may be provided for each electric storage cell in the case. The structure of the face F**1** of each case is not limited to the above structure, and can be altered when appropriate. For example, a pressure regulation valve for vacuum may be provided at the sealing hole **310** of each of the cases **301**, **302**, and the interiors of the cases **301**, **302** may be vacuumized. Further, at least one of a pressure regulation hole and a gas emission valve may be provided on the face F**2** of each of the cases **301**, **302**.

[0037] Each of the cell group G.sub.A and the cell group G.sub.B includes a plurality of electric storage cells that is connected in the X-direction. More specifically, the cell group G.sub.A includes a cell coupling body **10** (first cell coupling body) and a cell coupling body **20** (second cell coupling body) that are electrically connected. The cell group G.sub.B includes a cell coupling body **30** (first cell coupling body) and a cell coupling body **40** (second cell coupling body) that are electrically connected. Each electric storage cell included in the cell coupling bodies is configured to be capable of storing electricity.

[0038] The cell coupling body **10** includes four electric storage cells **11** to **14** that are arrayed in the X-direction, and three connection portions **2** that electrically connect the electric storage cells **11** to **14**. The electric storage cells **11** to **14** are connected on a line in the X-direction within the case **301**. The cell coupling body **20** includes four electric storage cells **21** to **24** that are arrayed in the X-direction, and three connection portions **2** that electrically connect the electric storage cells **21** to **24**. The electric storage cells **21** to **24** are connected on a line in the X-direction within the case **301**. In this way, the cell coupling body **10** and the cell coupling body **20** are disposed parallel in the X-direction. In the embodiment, the cell coupling bodies **10**, **20** are disposed such that the positions of the electric storage cells and the connection portions are arranged. The connection terminal T**1** is provided at a +X-side end portion (electric storage cell **11**) of the cell coupling body **10**, and the connection terminal T**2** is provided at a +X-side end portion (electric storage cell **21**) of the cell coupling body **20**. The electric storage cells **11**, **21** are electrically connected to the electrode tabs **321**, **322** (face F**1**) of the case **301** through the connection terminals T**1**, T**2**, respectively. A −X-side end portion (electric storage cell **14**) of the cell coupling body **10** and a −X-side end portion (electric storage cell **24**) of the cell coupling body **20** are electrically connected, for example, through a U-shaped connection portion **3**, within the case **301**.

[0039] The cell coupling body **30** includes four electric storage cells **31** to **34** that are arrayed in the

X-direction, and three connection portions **2** that electrically connect the electric storage cells **31** to **34**. The electric storage cells **31** to **34** are connected on a line in the X-direction within the case **302**. The cell coupling body **40** includes four electric storage cells **41** to **44** that are arrayed in the X-direction, and three connection portions **2** that electrically connects the electric storage cells **41** to **44**. The electric storage cells **41** to **44** are connected on a line in the X-direction within the case **302**. In this way, the cell coupling body **30** and the cell coupling body **40** are disposed parallel in the X-direction. In the embodiment, the cell coupling bodies **30**, **40** are disposed such that the positions of the electric storage cells and the connection portions are arranged. The connection terminal T**3** is provided at a +X-side end portion (electric storage cell **31**) of the cell coupling body **30**, and the connection terminal T**4** is provided at a +X-side end portion (electric storage cell **41**) of the cell coupling body **40**. The electric storage cells **31**, **41** are electrically connected to the electrode tabs **321**, **322** (face F**1**) of the case **302** through the connection terminals T**3**, T**4**, respectively. A −X-side end portion (electric storage cell **34**) of the cell coupling body **30** and a −X-side end portion (electric storage cell **44**) of the cell coupling body **40** are electrically connected, for example, through a U-shaped connection portion **3**, within the case **302**.

[0040] The connection portion **3** of each of the cell groups G.sub.A, G.sub.B may be an integrally molded article, or may be a composite of a plurality of separately molded components. For example, the connection portion **3** may be formed by connecting a protrusion portion **144**B (FIG. **2**) that protrudes from the electric storage cell **14** or **34** and a protrusion portion **144**B (FIG. **2**) that protrudes from the electric storage cell **24** or **44** through a conductive material (joist portion). The protrusion portion **144**B will be described later.

[0041] In the embodiment, the cell coupling bodies **10**, **20**, **30**, **40** basically have the same configuration. Hereinafter, the electric storage cells **11** to **14**, **21** to **24**, **31** to **34**, **41** to **44** are referred to as “electric storage cells **1**”, when they are not distinguished.

[0042] FIG. **2** is a diagram for describing the configuration of each cell coupling body. As shown in FIG. **2**, each cell coupling body includes four electric storage cells **1**. Moreover, the connection portion **2** is provided between adjacent electric storage cells **1**, and the connection portion **2** electrically connects the electric storage cells **1**. Each cell coupling body is configured such that the electric storage cell **1** and the connection portion **2** are alternately arrayed. In each cell coupling body, the electric storage cells **1** are connected to each other through the connection portion **2**. The stiffness of the connection portion **2** is lower than the stiffness of the electric storage cell **1**. The electric storage cells **11** to **14**, **21** to **24**, **31** to **34**, **41** to **44** are constituted by the same electric storage cell **1**. Since the cell coupling bodies **10**, **20**, **30**, **40** are formed using the common electric storage cell **1**, the electric storage device **100** is easily produced, and the production cost can be reduced.

[0043] In the embodiment, the electric storage cell **1** is a laminate cell that includes one or more wound bodies. In the laminate cell, one or more wound bodies that function as electrode bodies is covered with a laminate exterior packaging body. For example, the wound body has a structure in which a positive electrode sheet and a negative electrode sheet are wound such that a separator is interposed. Each of the positive electrode sheet and the negative electrode sheet includes an electrode foil and an active material layer.

[0044] The structures of the electric storage cell **1** and the connection portion **2** will be described below using a sectional view (an X-Y sectional view of the periphery of the connection portion **2**) in FIG. **2**. The electric storage cell **1** includes two wound bodies **110**A, **110**B, spacers **120**A, **120**B, terminal members **130**A, **130**B, and covers **150**A, **150**B.

[0045] The wound bodies **110**A, **110**B include coated portions **111**A, **111**B, electrode tabs **112**A, **112**B, and electrode tabs **113**A, **113**B, respectively. Each of the coated portions **111**A, **111**B is a region where the active material layer of the electrode foil of the positive electrode sheet or the negative electrode sheet is provided. Each of the electrode tabs **112**A, **112**B, **113**A, **113**B is a region where the electrode foil is exposed on the positive electrode sheet or the negative electrode sheet (an uncoated portion where the active material layer is not provided). The electrode tabs **112**A, **112**B are positioned at +X-side end portions of the wound bodies **110**A, **110**B, respectively. The electrode tabs **113**A, **113**B are positioned at −X-side end portions of the wound bodies **110**A, **110**B, respectively.

[0046] The electrode tab **112**A and the electrode tab **112**B are disposed so as to overlap in the Y-direction, and the spacer **120**A and the terminal member **130**A are provided between the electrode tab **112**A and the electrode tab **112**B. The electrode tab **113**A and the electrode tab **113**B are disposed so as to overlap in the Y-direction, and the spacer **120**B and the terminal member **130**B are provided between the electrode tab **113**A and the electrode tab **113**B.

[0047] Each of the spacers **120**A, **120**B contains an insulating material (for example, a synthetic resin), and has an insulation property. Each of the spacers **120**A, **120**B has a shape in which the dimension in the Y-direction is larger at a position farther from the coated portions **111**A, **111**B. The terminal member **130**A is connected to a +X-side end face of the spacer **120**A. The terminal member **130**B is connected to a −X-side end face of the spacer **120**B. Each of the terminal members **130**A, **130**B contains a conducting material (for example, a metal such as aluminum or copper), and has a conductive property. The wound body **110**A and the wound body **110**B are joined to each other through the terminal members **130**A, **130**B (for example, by laser welding).

[0048] Each of current collecting terminals **140**A, **140**B is a component that constitutes a part of the connection portion **2**. The current collecting terminals **140**A, **140**B include support portions **142**A, **142**B and protrusion portions **144**A, **144**B, respectively. One of the current collecting terminals **140**A, **140**B functions as a positive electrode current collecting terminal, and the other of current collecting terminals **140**A, **140**B functions as a negative electrode current collecting terminal. As an example, the positive electrode current collecting terminal is formed of aluminum, and the negative electrode current collecting terminal is formed of copper.

[0049] Each of the current collecting terminals **140**A, **140**B, which is formed in an L-shape, may be formed by joining two plate members that are separately molded, or may be formed by bending process in an integral state. The support portion **142**A is joined to a +X-side end face of the terminal member **130**A (for example, by laser welding). The support portion **142**B is joined to a −X-side end face of the terminal member **130**B (for example, by laser welding).

[0050] The cover **150**A covers a +X-side end portion (including the electrode tabs **112**A, **112**B) of the electric storage cell **1**. A through-hole h**1** for the protrusion portion **144**A is provided on the cover **150**A. The protrusion portion **144**A passes through the through-hole h**1**, and protrudes to the +X-side of the electric storage cell **1**. Further, the cover **150**B covers a −X-side end portion (including the electrode tabs **113**A, **113**B) of the electric storage cell **1**. A through-hole h**2** for the protrusion portion **144**B is provided on the cover **150**B. The protrusion portion **144**B passes through the through-hole h**2**, and protrudes to the −X-side of the electric storage cell **1**.

[0051] At the connection portion **2**, the protrusion portion **144**A that protrudes from one electric storage cell **1** of the two adjacent electric storage cells **1** and the protrusion portion **144**B that protrudes from the other electric storage cell **1** are joined (for example, by laser welding). The welded portion may be protected by a tape or the like. A laminate exterior packaging body **160** is provided on surfaces of the two wound bodies **110**A, **110**B. The laminate exterior packaging body **160** is a laminate film, for example, and is provided on a surface of the electric storage cell **1**.

[0052] FIG. **3** is a sectional view taken along line III-III in FIG. **1**. Each of the cases **301** to **303** shown in FIG. **1** has a U-shaped section shown in FIG. **3**, as a section (Y-Z section) orthogonal to the X-direction. Further, in each of the cases **301**, **302**, a region R exists between the +Z-side face of each electric storage cell and an inner face (top face) of the case. At the region R, at least one of a heat management system (for example, a heater and/or a temperature sensor), a gas emission system (for example, a gas flow passage and/or a pressure sensor), a flexible printed circuit board (FPC), and a wire that is connected to the connector **330** may be provided. Devices and/or sensors provided at the region R may be connected to the connector **330**. Each electric storage cell housed

in cases **301**, **302** may be caused to adhere to the inner face of the case by an adhesive agent.
[0053] The above configuration is just an example of the configuration of the electric storage cell **1**, and can be altered when appropriate. For example, the number of wound bodies that are included in the electric storage cell **1** is not limited to two, and may be one, or may be three or more. Further, as the electrode body, a laminated body (for example, a laminated body in which a positive electrode sheet and a negative electrode sheet are laminated such that a separator is interposed) may be employed instead of the wound body. Further, for example, an insulating layer containing a resin such as polyethylene terephthalate (PET) may be provided on the inner face of each case.
[0054] FIG. **4** is a diagram for describing operations and effects of the electric storage device **100**. In a mode in which the electrically connected cell coupling bodies **10**, **20** are housed in a case **400** having a bottomed tubular shape and including an opening portion on the +X-side end face as shown in a reference example in FIG. **4**, there can be a problem in that it is difficult to insert the cell coupling bodies **10**, **20** into the case **400**. Further, in this reference example, for causing the interior of the case **400** to be in a sealed state, a lid member for closing the opening portion of the case **400** is necessary. In contrast, in the electric storage device **100** shown in FIG. **1** to FIG. **3**, the opening portion of the case **301** that houses the cell group G.sub.A is closed by the case **302** that houses another cell group (cell group G.sub.B), and therefore, it is not necessary to separately provide the lid member for closing the opening portion of the case **301**. Thereby, the increase in the number of components is restrained. Further, the case **301** includes the opening portion having a size that allows the cell group G.sub.A to easily pass (specifically, a size equivalent to the size of the face F**3** on the opposite side). Since the entrance (opening) of the case **301** is wide, the cell group G.sub.A is easily put in the case **301**. Moreover, the case **302** has the same shape and dimensions as the case **301**, and therefore, the opening portion of the case **301** can be closed by the face F**3** of the case **302**. Furthermore, the case **302** includes the opening portion having a size that allows the cell group G.sub.B to easily pass (specifically, a size equivalent to the size of the face F**3** on the opposite side). Since the entrance (opening) of the case **302** is wide, the cell group G.sub.B is easily put in the case **302**. Moreover, the case **303** has the same shape and dimensions as the case **302**, and therefore, the opening portion of the case **302** can be closed by the face F**3** of the case **303**.
[0055] The number of cases that house cell groups is not limited to two, and may be three or more. FIG. **5** is a diagram showing a first modification of the electric storage device **100**.
[0056] As shown in FIG. **5**, an electric storage device **100**A according to the first modification includes N cases (cases **300-1**, **300-2**, . . . , **300**-N) and N cell groups (cell groups G-**1**, G-**2**, . . . , G-N). Specifically, the electric storage device **100**A includes cases **300-1** to **300**-N and cell groups G-**1** to G-N that are housed in the cases **300-1** to **300**-N, respectively. Each of the cell groups G-**1** to G-N has the same configuration as the above-described cell group G.sub.A (see FIG. **1** to FIG. **3**). N may be 3 or more and less than 20, or may be 20 or more.
[0057] Each of the cases **300-1** to **300**-N has the same shape as the above-described case **301** (see FIG. **4**). As shown in FIG. **5**, the cases **300-1** to **300**-N are stacked in the Y-direction. The electric storage device **100**A further includes a constraining tool that constrains the cases (laminated body) in the Y-direction. The constraining tool includes pressing plates **501**, **502**, connection shafts **503**, **504**, and bolts P**1** to P**4**. The pressing plate **501** is coupled to the connection shafts **503**, **504** by the bolts P**1**, P**2**, respectively. Further, the pressing plate **502** is coupled to the connection shafts **503**, **504** by the bolts P**3**, P**4**, respectively. By tightening the bolts P**1** to P**4**, the pressure (constraining force) in the Y-direction is given to the laminated body (cases **300-1** to **300**-N). The interior of each case may be vacuumized, and the constraining pressure may be increased.
[0058] The opening portion of each of the cases **300-1** to **300**-N is closed. The opening portion of the case **300**-N that is at the outermost position on the +Y-side among the cases **300-1** to **300**-N is closed by the pressing plate **501** (for example, a flat plate member), and the opening portion of each of the other cases is closed by the case that is at the adjacent position (on the +Y-side). The

pressing plate **501** functions as a lid member that closes the opening portion of the case **300**-N. The constraining force may be given to the laminated body (cases **300-1** to **300**-N), such that the pressing plate **501** presses the case **300**-N hard, and thereby, the airtightness of each case may be increased. In order that the face pressure by the constraining tool is uniformly given to the case **300**-N, an elastic body and/or an adhesive agent may be provided between the case **300**-N and the pressing plate **501**. Each case may be constrained by a band (for example, a rubber band), instead of the bolts (fastening).
[0059] The number of cell coupling bodies that are housed in the case is not limited to two, and may be an arbitrary number. The number of cell coupling bodies that are housed in the case may be three or more, or may be one. FIG. **6** is a diagram showing a second modification of the electric storage device **100**.
[0060] As shown in FIG. **6**, an electric storage device **100**B according to the second modification includes a case **301**A (first case) that houses a cell group Gc (first cell group), and a case **302**A (second case) that houses a cell group GD (second cell group). Each of the cell groups Gc, Gp is one cell coupling body that has the configuration shown in FIG. **2**.
[0061] Each of the cases **301**A, **302**A basically has the same shape as the above-described case **301** (see FIG. **4**). However, in the case **301**A, an electrode tab **321**A is provided on the face F**1**, and an electrode tab **322**A is provided on the face F**2**. Moreover, an electric storage cell **1** positioned at one X-directional end (+X-side) of the cell coupling body that is housed in the case **301**A includes a connection terminal T**1**A that is electrically connected to the electrode tab **321**A. Further, an electric storage cell **1** positioned at the other X-directional end (−X-side) of the cell coupling body that is housed in the case **301**A includes a connection terminal T**2**A that is electrically connected to the electrode tab **322**A. Further, in the case **302**A, an electrode tab **321**B is provided on the face F**1**, and an electrode tab **322**B is provided on the face F**2**. Moreover, an electric storage cell **1** positioned at the one X-directional end (+X-side) of the cell coupling body that is housed in the case **302**A includes a connection terminal T**1**B that is electrically connected to the electrode tab **321**B. Further, an electric storage cell **1** positioned at the other X-directional end (−X-side) of the cell coupling body that is housed in the case **302**A includes a connection terminal T**2**B that is electrically connected to the electrode tab **322**B.
[0062] In the electric storage device **100**B, the face F**3** of the case **302**A closes the opening portion of the case **301**A. Thereby, the cell group Gc is protected while being surrounded by the cases **301**A, **302**A. Further, a lid member may be provided on the +Y-side of the case **302**A, and the opening portion of the case **302**A may be closed by the lid member. In this configuration, the cell group Gp is protected while being surrounded by the case **302**A and the lid member. The lid member may be another case that houses the cell group, or may be a lid member that has a flat plate shape.
[0063] The configuration of each cell coupling body is not limited to the configuration shown in FIG. **2**. Each cell coupling body may include electric storage cells having different dimensions, and may include electric storage cells having different shapes. The number of the electric storage cells that are included in each cell coupling body may be less than 4, may be 5 or more and 19 or less, or may be 20 or more.
[0064] All cases that house cell groups do not need to have the same shape. Further, all cases that house cell groups do not need to be sealed. For example, the electric storage device **100** shown in FIG. **1** may be altered to a configuration in which the case **302** is opened in a state where the case **302** houses the cell group G.sub.B, by removing the case **303** and the biasing device **50** from the electric storage device **100**. Since the case **302** is opened, the heat dissipation of the cell group G.sub.B can be enhanced.
[0065] The above-described electric storage devices **100**, **100**A, **100**B can be equipped in a movable body, for example. Examples of the movable body include a vehicle (a battery electric vehicle, hybrid electric vehicle, and the like), a conveyance (a ship, an airplane, and the like) other

than the vehicle, a movable machine (an agricultural machine, a construction machine, and the like), and an unmanned movable body (an unmanned carrier, a robot, and the like). The electric storage device is used for an arbitrary purpose, and may be a stationary electric storage device.
[0066] Various characteristics relevant to the above-described electric storage devices (characteristics described in the embodiment and the modifications) may be carried out while being arbitrarily combined.
[0067] It should be understood that the embodiment and modifications disclosed herein are examples and are not limitative in all respects. It is intended that the scope of the present disclosure is shown not by the above descriptions of the embodiment and the modifications but by the claims and includes all alterations within meanings and ranges equivalent to the claims.

Claims

1. An electric storage device comprising: a first case that houses a first cell group; and a second case that houses a second cell group, wherein: each of the first cell group and the second cell group includes a plurality of electric storage cells that is connected in a first direction; the first case includes an opening portion at one end in a second direction orthogonal to the first direction; and the first case and the second case are disposed such that the second case closes the opening portion of the first case.
2. The electric storage device according to claim 1, wherein: each of the first case and the second case has a rectangular parallelepiped shape in which an opening portion is provided at one end in the second direction; each of the first case and the second case includes a first face and a second face that are positioned at both ends in the first direction, and a third face, a fourth face, and a fifth face that extend in the first direction; the third face is positioned on an opposite side of the opening portion in the second direction; the opening portion of the first case has a larger area than each of the first face and the second face of the first case; and the third face of the second case closes the opening portion of the first case.
3. The electric storage device according to claim 2, wherein: the first case and the second case are joined such that the first case is sealed in a state where the first case houses the first cell group; and the electric storage device further comprises a lid member that closes the opening portion of the second case such that the second case is sealed in a state where the second case houses the second cell group.
4. The electric storage device according to claim 1, further comprising a mechanism that generates force by which the second case is pressed against the first case.
5. A vehicle comprising the electric storage device according to claim 1.